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Assignment 7: Implementation of security policies with GPO (Group Policy Objects)

<https://github.com/elovric13/eutech/tree/implementation-of-security-policies-with-GPO-elovric13>

1. Install and configure a Domain Controller in Windows Server

Windows Server 2022 was installed in a virtual machine. After installation, the Active Directory Domain Services (AD DS) role was added through the Server Manager. The server was then promoted to a Domain Controller and a new forest was created with domain name **elovric.local**.

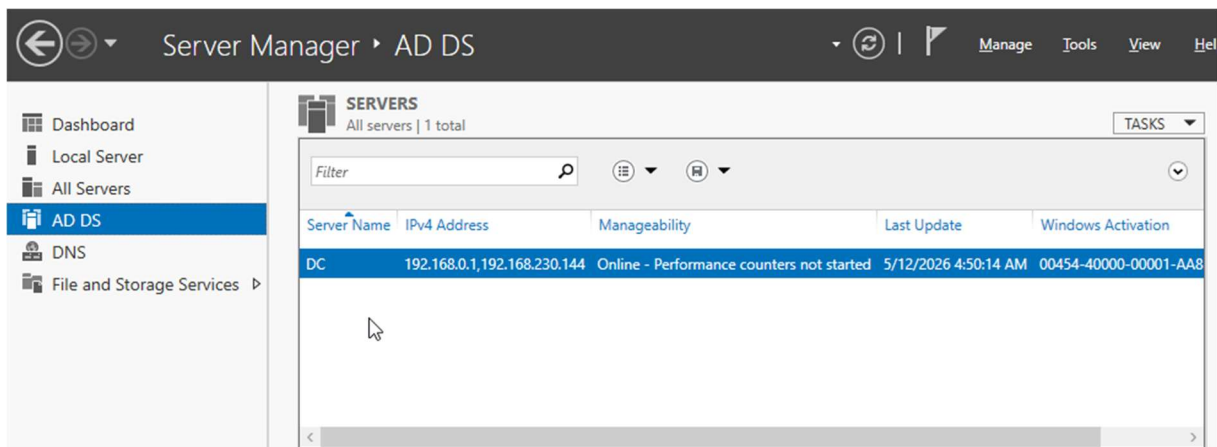


Figure 1: AD DS role

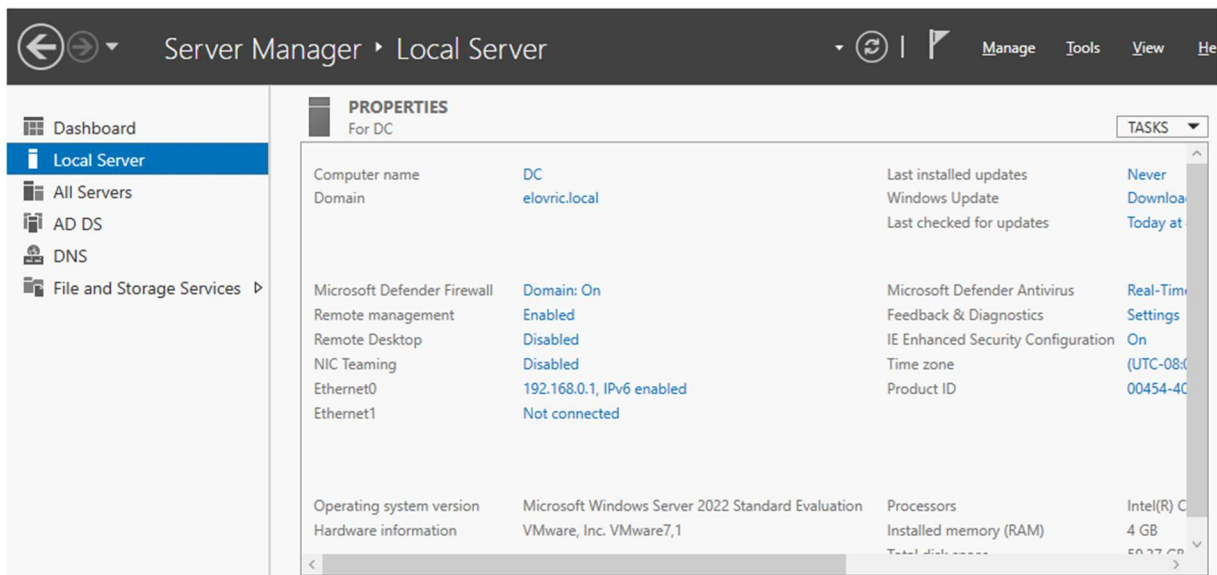


Figure 2: Domain Controller configuration

2. Create at least two Organizational Units (OUs) with different types of users

Two Organizational Units were created: **Students** and **Teachers**. Inside each OU, test user account was created to simulate real users belonging to each group.

Organizational Unit	Username
Students	student1
Teachers	teacher1

```
PS C:\Users\Administrator> Get-ADUser -Filter * | Select Name, DistinguishedName

Name                DistinguishedName
----                -
Administrator      CN=Administrator,CN=Users,DC=elovric,DC=local
Guest               CN=Guest,CN=Users,DC=elovric,DC=local
krbtgt              CN=krbtgt,CN=Users,DC=elovric,DC=local
student1            CN=student1,OU=Students,DC=elovric,DC=local
teacher1            CN=teacher1,OU=Teachers,DC=elovric,DC=local
```

Figure 3: Students and Teachers OUs and user accounts

3. Apply security policies with GPO

Group Policy Objects were created and linked to each Organizational Unit to enforce different security restrictions. The following policies were applied:

- Restricting access to the Control Panel

A GPO was created and linked to the Students OU to block access to the Control Panel and Settings. This was configured under User Configuration > Administrative Templates > Control Panel > Prohibit access to Control Panel and PC Settings, set to Enabled.

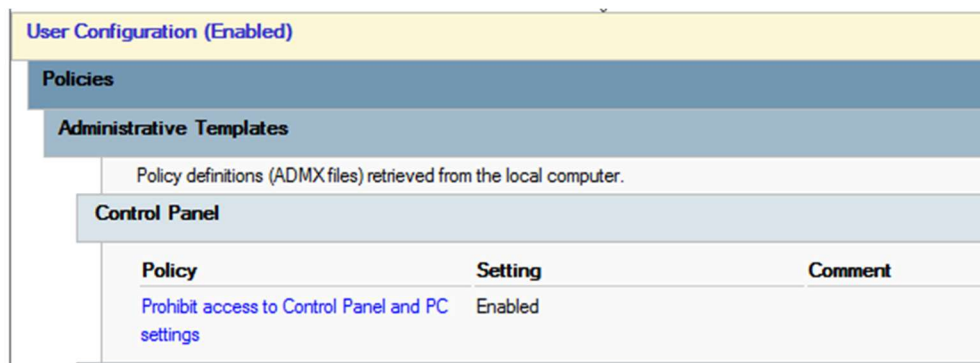


Figure 4: GPO setting for Control Panel restriction

- Disabling software installation

Software installation was disabled for the Students OU by enabling the policy under User Configuration > Administrative Templates > Windows Components > Windows Installer > Always install with elevated privileges, set to Enabled.

Windows Components/Windows Installer		
Policy	Setting	Comment
Always install with elevated privileges	Enabled	
This policy setting must be set for the machine and the user to be enforced.		

Figure 5: GPO setting for disabling software installation

- Enforcing strong password policies

A password policy was enforced at the domain level through the Default Domain Policy. The following settings were configured under Computer Configuration > Windows Settings > Security Settings > Account Policies > Password Policy:

- Minimum password length: 8 characters
- Password complexity requirements: Enabled
- Maximum password age: 30 days
- Minimum password age: 1 day
- Enforce password history: 24 passwords remembered

Policies	
Windows Settings	
Security Settings	
Account Policies/Password Policy	
Policy	Setting
Enforce password history	24 passwords remembered
Maximum password age	30 days
Minimum password age	1 days
Minimum password length	8 characters
Password must meet complexity requirements	Enabled
Store passwords using reversible encryption	Disabled

Figure 6: Password policy configuration in Default Domain Policy

4. Log in with test users from each group and verify the restrictions applied

To verify the GPO policies, gpupdate /force was run on the machine to apply the policies immediately. Test users from both OUs were then used to log in and check the applied restrictions.

When logged in as student1, access to the Control Panel was blocked. Attempts to install software also failed. When logged in as teacher1, both the Control Panel and software installation were accessible, confirming the policies were correctly scoped to the Students OU only. When trying to change a password that doesn't meet the requirements, the change was denied.

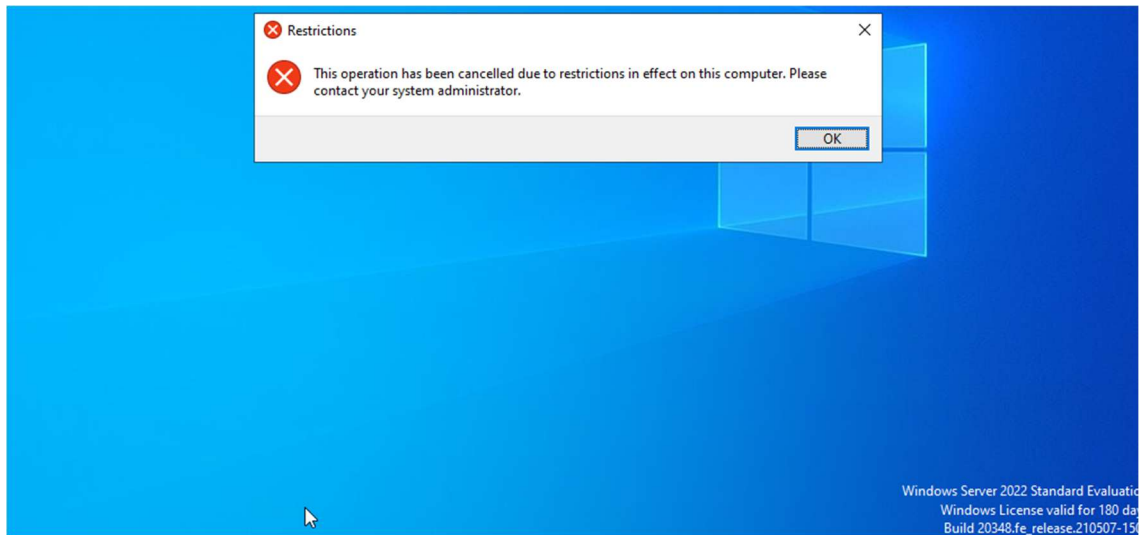


Figure 7: Control Panel access blocked on student account

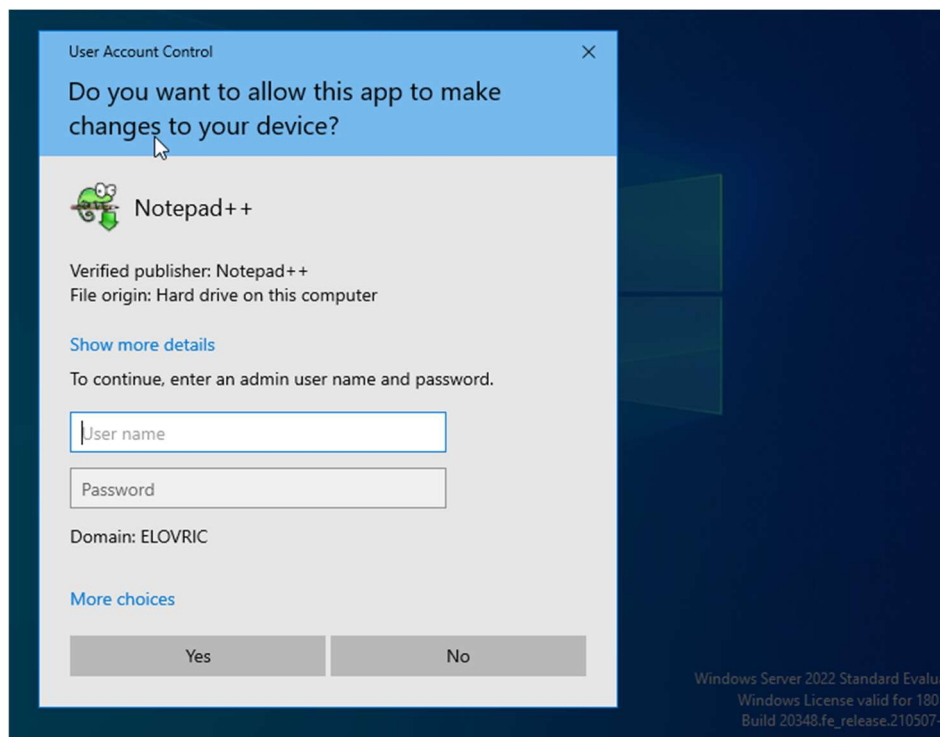


Figure 8: Software installation blocked on student account

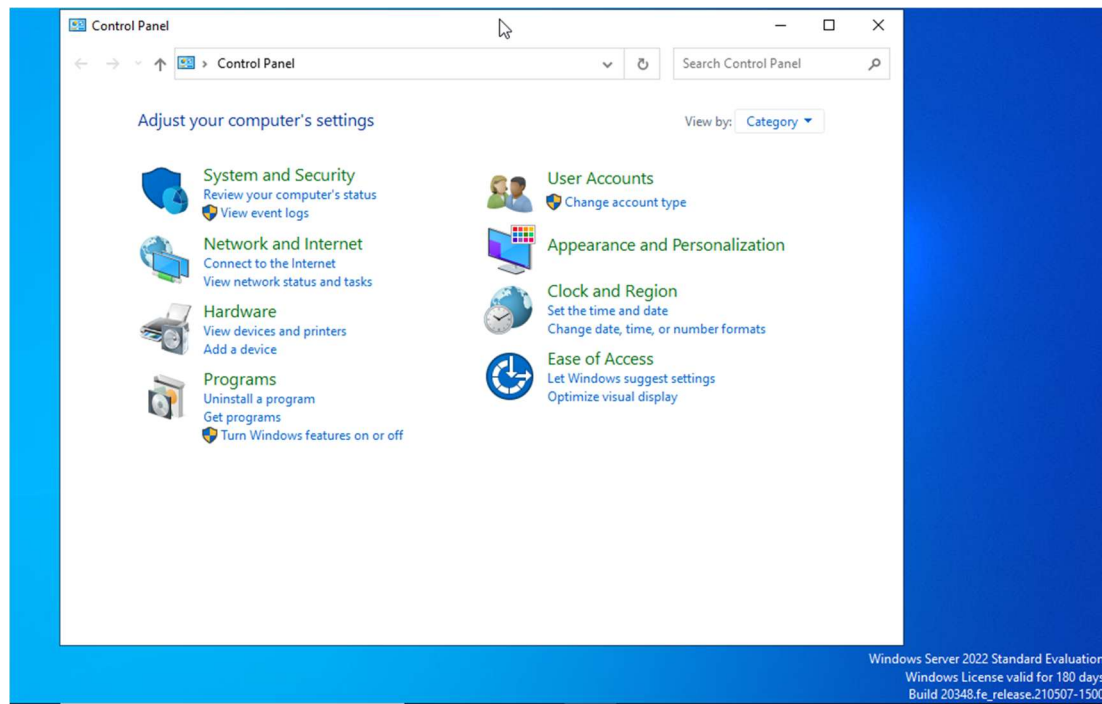


Figure 9: Control Panel accessible on teacher account

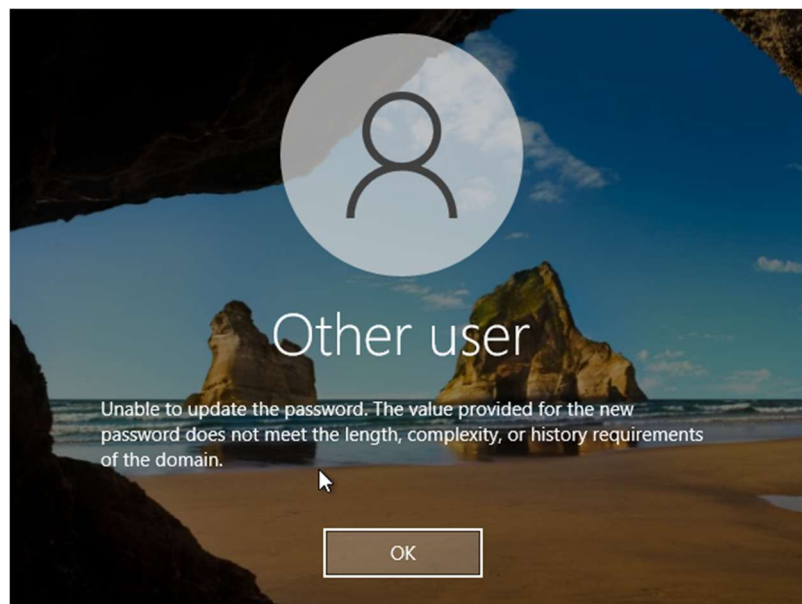


Figure 10: Password change failed

5. Conclusion

This task demonstrated how Group Policy Objects in a domain can be used to enforce different security restrictions for different groups of users. By organizing users into Organizational Units and linking GPOs to each OU, it is possible to apply targeted policies without affecting other users on the domain. The tests confirmed that the restrictions were correctly applied to the Students OU while the Teachers OU remained unaffected.